STAT 140 Midterm Review

Design of Experiments

- Running example: crop yield under different fertilizer treatments

For each unit i (a plot of land), define potential outcomes $Y_i(1)$ (yield under new fertilizer) and $Y_i(0)$ (yield under standard fertilizer). The observed outcome is $Y_i^{\text{obs}} = W_i Y_i(1) + (1 - W_i) Y_i(0)$. The fundamental problem of causal inference is that we observe at most one potential outcome per unit.

1 Completely Randomized Design (CRE)

Assign N_T units to treatment and N_C units to control ($N = N_T + N_C$), chosen uniformly at random from all $\binom{N}{N_T}$ valid assignments.

Assignment Mechanism

$$|W^+| = \binom{N}{N_T}, \qquad P(W) = \frac{1}{\binom{N}{N_T}}, \qquad P(W_i = 1) = \frac{N_T}{N}$$

Fertilizer Example

Randomly select half the plots to receive the new fertilizer; the other half receive the standard fertilizer.

Key Insight

Randomization guarantees **unbiasedness** of the difference-in-means estimator, but may produce covariate imbalance by chance (e.g. too many south-facing plots in one group).

Fisherian Inference for CRE

Sharp null: $H_0 : Y_i(1) = Y_i(0)$ for all i. Under H_0, each unit's outcome is fixed regardless of assignment. Test statistic:

$$T_{\text{dm}} = \bar{Y}_T^{\text{obs}} - \bar{Y}_C^{\text{obs}} = \frac{1}{N_T} \sum_{i:\, W_i = 1} Y_i^{\text{obs}} - \frac{1}{N_C} \sum_{i:\, W_i = 0} Y_i^{\text{obs}}$$

Note

Step 1. Compute $T^{\text{obs}} = \bar{Y}_T^{\text{obs}} - \bar{Y}_C^{\text{obs}}$ under the actual assignment.
Step 2. For each of the $\binom{N}{N_T}$ assignments $W \in W^+$, relabel the observed outcomes and recompute $T(W)$.
Step 3. p-value $= \frac{1}{|W^+|} \#\big\{W \in W^+ : |T(W)| \geq |T^{\text{obs}}|\big\}$.
When $|W^+|$ is large, approximate by Monte Carlo sampling.

Neymanian Inference for CRE

Estimand: $\tau = \frac{1}{N} \sum_{i=1}^{N} \big(Y_i(1) - Y_i(0)\big)$. Estimator: $\hat{\tau} = \bar{Y}_T^{\text{obs}} - \bar{Y}_C^{\text{obs}}$, unbiased with $\mathbb{E}[\hat{\tau}] = \tau$.

Exact variance:

$$\mathrm{Var}(\hat{\tau}) = \frac{S_1^2}{N_T} + \frac{S_0^2}{N_C} - \frac{S_\tau^2}{N}$$

where S_1^2, S_0^2 are the finite-population variances of $Y_i(1)$ and $Y_i(0)$, and $S_\tau^2 = \frac{1}{N-1}\sum_i (Y_i(1) - Y_i(0) - \tau)^2$.

S_τ^2 is Unobservable

We never see both $Y_i(1)$ and $Y_i(0)$ for the same unit. Conservative estimator (drop S_τ^2/N):

$$\widehat{\mathrm{Var}}(\hat{\tau}) = \frac{s_1^2}{N_T} + \frac{s_0^2}{N_C} \geq \mathrm{Var}(\hat{\tau})$$

Tight when treatment effects are constant: $Y_i(1) - Y_i(0) = \tau$ for all i.

2 Matched Pair Design

Units are indexed with a double subscript (i, p) where $i \in \{1, 2\}$ is the within-pair index and $p \in \{1, \ldots, P\}$ is the pair. We have $N = 2P$ total units. Within each pair, one unit is treated and one is control, chosen by an independent coin flip.

Assignment Mechanism

$$|W^+| = 2^P, \qquad P(W) = 2^{-P}, \qquad P(W_{i,p} = 1) = \tfrac{1}{2}$$

The observed outcome: $Y_{i,p}^{\mathrm{obs}} = W_{i,p} Y_{i,p}(1) + (1 - W_{i,p}) Y_{i,p}(0)$.

Fertilizer Example

Pair plots that share similar soil quality and sun exposure. Within each pair, randomly assign one plot to the new fertilizer.

Why Pairing Helps

Within each pair, exactly one unit is treated and one is control, so the pair-level observed difference $Y_{p,t}^{\mathrm{obs}} - Y_{p,c}^{\mathrm{obs}}$ estimates the pair-level effect τ_p directly. Between-pair variation (e.g. north vs. south field) cancels out entirely and does not inflate $\mathrm{Var}(\hat{\tau})$.

Three Levels of Causal Estimands

- Unit-level: $\tau_{i,p} = Y_{i,p}(1) - Y_{i,p}(0)$ (unobservable)
- Pair-level: $\tau_p = \frac{1}{2}\sum_{i \in \text{pair } p} \big(Y_{i,p}(1) - Y_{i,p}(0)\big)$
- Average (ATE): $\tau = \frac{1}{N}\sum_{i,p} \big(Y_{i,p}(1) - Y_{i,p}(0)\big) = \frac{1}{P}\sum_{p=1}^{P} \tau_p$

Fisherian Inference for Matched Pair

Sharp null: $H_0 : Y_{i,p}(1) = Y_{i,p}(0)$ for all (i, p). Let $d_p = Y_{p,t}^{\mathrm{obs}} - Y_{p,c}^{\mathrm{obs}}$ be the within-pair observed difference. Test statistics:

$$T_{\mathrm{dm}} = \bar{d} = \frac{1}{P}\sum_{p=1}^{P} d_p \qquad \text{or} \qquad T_{\mathrm{paired}} = \frac{\bar{d}}{s_d/\sqrt{P}}$$

where $s_d^2 = \frac{1}{P-1}\sum_{p=1}^{P}(d_p - \bar{d})^2$.

Under H_0, swapping the treatment and control labels within pair p changes $d_p \to -d_p$. This gives 2^P valid sign-flip assignments.

Note

Step 1. Compute $T^{\text{obs}} = \bar{d}$ from observed pair differences $d_1, \ldots, d_P$.
Step 2. For each of the 2^P sign-flip assignments, flip any subset of signs to get d_p^*, then recompute $\bar{d}^*$.
Step 3. p-value $= \frac{1}{2^P}\#\{\text{assignments} : |\bar{d}^*| \geq |T^{\text{obs}}|\}$.
Re-randomization only swaps labels within pairs – it never moves a unit from one pair to another. The reference set has 2^P elements, not $\binom{N}{N/2}$.

Neymanian Inference for Matched Pair

Pair-level estimator: $\hat{\tau}_p = Y_{p,t}^{\text{obs}} - Y_{p,c}^{\text{obs}} = d_p$, which is unbiased for τ_p.

Overall ATE estimator and its variance:

$$\hat{\tau} = \frac{1}{P}\sum_{p=1}^{P}\hat{\tau}_p = \bar{d}, \qquad \widehat{\text{Var}}(\hat{\tau}) = \frac{1}{P(P-1)}\sum_{p=1}^{P}(\hat{\tau}_p - \hat{\tau})^2 = \frac{s_d^2}{P}$$

This variance estimator is **exact** (not conservative): s_d^2 is directly observed from data.

The exact (unobservable) variance involves within-pair terms $S_c^2(p)$, $S_t^2(p)$, and $S_{ct}^2(p)$:

$$\text{Var}(\hat{\tau}_p) = S_c^2(p) + S_t^2(p) - \frac{S_{ct}^2(p)}{2}$$

where $S_{ct}^2(p)$ is the covariance of unit-level effects within pair p. This term is unobservable (it requires both $Y_{i,p}(1)$ and $Y_{i,p}(0)$), but good pairing makes $S_{ct}^2(p)$ large and positive, reducing variance.

3 Multi-Treatment CRE

We have N units and $J \geq 2$ treatments. A pre-specified number N_j of units is assigned to treatment j, with $\sum_{j=1}^{J} N_j = N$. Each unit i has J potential outcomes $Y_i(j)$ for $j = 1, \ldots, J$. The observed outcome is:

$$Y_i^{\text{obs}} = \sum_{j=1}^{J}\mathbf{1}(W_i = j)\, Y_i(j)$$

Assignment Mechanism

$$|W^+| = \frac{N!}{N_1!\, N_2! \cdots N_J!}, \qquad P(W) = \frac{1}{|W^+|} \text{ for } W \in W^+$$

Fertilizer Example

Compare three fertilizer types: standard ($j = 1$), low-dose new ($j = 2$), high-dose new ($j = 3$). Each plot is randomly assigned to exactly one, with $N_1 = N_2 = N_3 = N/3$.

Causal Estimands

Unit-level effect of j vs. j': $\tau_i(j, j') = Y_i(j) - Y_i(j')$ (unobservable).

Average causal effect (ATE): $\tau(j, j') = \frac{1}{N}\sum_{i=1}^{N} \tau_i(j, j') = \bar{Y}(j) - \bar{Y}(j')$, where $\bar{Y}(j) = \frac{1}{N}\sum_i Y_i(j)$ is the j-th column mean of the science table.

Fisherian Inference for Multi-Treatment CRE

Sharp null: $H_0 : Y_i(1) = Y_i(2) = \cdots = Y_i(J)$ for all i. Under H_0, the entire science table is known. The natural test statistic is the ANOVA F-statistic:

$$F = \frac{MS_{\text{Treatment}}}{MS_{\text{Residual}}} = \frac{SS_{\text{Treatment}}/(J-1)}{SS_{\text{Residual}}/(N-J)}$$

where:

$$SS_{\text{Treatment}} = \sum_{j=1}^{J} N_j \big(\bar{Y}_j^{\text{obs}} - \bar{Y}^{\text{obs}}\big)^2, \qquad SS_{\text{Residual}} = \sum_{j=1}^{J} \sum_{i:\, W_i = j} \big(Y_i^{\text{obs}} - \bar{Y}_j^{\text{obs}}\big)^2$$

Note: ANOVA Table

Source	df	SS	MS
Treatments	$J-1$	$SS_{\text{Treatment}}$	$SS_{\text{Treatment}}/(J-1)$
Residuals	$N-J$	SS_{Residual}	$SS_{\text{Residual}}/(N-J)$
Total	$N-1$	SS_{Total}	

$F \approx 1$ under H_0 (between-group variation $\approx$ within-group noise). Large F is evidence against H_0.
For a pairwise contrast, one can also use $T = \sum_j \lambda_j \bar{Y}_j^{\text{obs}}$ with $\sum_j \lambda_j = 0$. In either case, the p-value is computed by re-permuting W over all $\frac{N!}{\prod N_j!}$ assignments.

Neymanian Inference for Multi-Treatment CRE

Estimator for $\tau(j, j')$: $\hat{\tau}(j, j') = \bar{Y}_j^{\text{obs}} - \bar{Y}_{j'}^{\text{obs}}$, unbiased for $\tau(j, j')$.

Exact variance:

$$\text{Var}\big(\hat{\tau}(j, j')\big) = \frac{S_j^2}{N_j} + \frac{S_{j'}^2}{N_{j'}} - \frac{S_{jj'}^2}{N}$$

where $S_j^2 = \frac{1}{N-1}\sum_i (Y_i(j) - \bar{Y}(j))^2$ and $S_{jj'}^2 = \frac{1}{N-1}\sum_i (\tau_i(j, j') - \tau(j, j'))^2$.

Conservative estimated variance (drop $S_{jj'}^2/N \geq 0$):

$$\widehat{\text{Var}}\big(\hat{\tau}(j, j')\big) = \frac{s_j^2}{N_j} + \frac{s_{j'}^2}{N_{j'}}$$

where $s_j^2 = \frac{1}{N_j - 1}\sum_{i:\, W_i = j} (Y_i^{\text{obs}} - \bar{Y}_j^{\text{obs}})^2$.

Key Insight

With J arms there are $\binom{J}{2}$ pairwise comparisons. Running them all inflates the family-wise error rate. If the ANOVA F-test rejects first, pairwise comparisons are better justified.

4 Crossover Design

Each participant i is observed over two periods $j \in \{1, 2\}$. The treatment indicator $W_{i,j} \in \{0, 1\}$ satisfies the crossover constraint:

$$W_{i,j=2} = 1 - W_{i,j=1}$$

so participants receive treatment and control in opposite periods. The randomness is entirely in which period each participant is treated first.

Each participant has four potential outcomes: $Y_{i,j}(0)$ and $Y_{i,j}(1)$ for $j \in \{1, 2\}$. Observed outcome: $Y_{i,j}^{\text{obs}} = W_{i,j} Y_{i,j}(1) + (1 - W_{i,j}) Y_{i,j}(0)$.

Assignment Mechanism (Bernoulli)

Each participant independently flips a fair coin for their period-1 assignment:

$$|W^+| = 2^N, \qquad P(W) = \left(\tfrac{1}{2}\right)^N$$

Fertilizer Example

Each plot is used across two growing seasons. Randomly assign which season gets the new fertilizer; the other season uses the standard fertilizer.

Key Insight

Each unit acts as its own control, eliminating between-unit variation entirely. Key assumption: **no carryover effect** and **no time effect** – the potential outcome in period 2 must not be influenced by treatment received in period 1, and the two periods must be otherwise exchangeable.

Four Levels of Causal Estimands

- Unit-period effect: $\tau_{i,j} = Y_{i,j}(1) - Y_{i,j}(0)$
- Period-averaged participant effect: $\tau_i = \frac{1}{2}(\tau_{i,1} + \tau_{i,2})$
- Population-average period effect: $\tau_j = \frac{1}{N} \sum_{i=1}^{N} \tau_{i,j}$
- Grand average: $\tau = \frac{1}{2}(\tau_{j=1} + \tau_{j=2})$

A key feature: under no carryover/time effects, participant i is observed under both conditions, so τ_i is directly estimable – something impossible in any between-person design.

Fisherian Inference for Crossover

For each participant i, define the within-person difference:

$$d_i = \begin{cases} Y_{i,j=1}(1) - Y_{i,j=2}(0) & \text{if } W_{i,1} = 1 \text{ (treatment first)} \\ Y_{i,j=2}(1) - Y_{i,j=1}(0) & \text{if } W_{i,1} = 0 \text{ (control first)} \end{cases}$$

In both cases, $d_i =$ (treatment period outcome) $-$ (control period outcome) for participant i. Test statistic:

$$T_{\text{paired}} = \frac{\bar{d}}{s_D/\sqrt{N}}, \qquad s_D^2 = \frac{1}{N-1} \sum_{i=1}^{N} (d_i - \bar{d})^2$$

Under $H_0 : Y_{i,j}(1) = Y_{i,j}(0)\ \forall\, i, j$, all potential outcomes are fixed. Re-randomizing means independently flipping each participant's sequence: 2^N valid sign-flip assignments.

Note

Step 1. Compute $T^{\text{obs}} = \bar{d}/(s_D/\sqrt{N})$ from observed within-person differences.
Step 2. For each of the 2^N sequence-flip assignments, flip the sign of any subset of the d_i and recompute T^*.
Step 3. p-value $= \Pr(|T^*| \geq |T^{\text{obs}}|)$ over all 2^N assignments.

Neymanian Inference for Crossover

Natural estimator for τ_i: $\hat{\tau}_i = d_i$ (unbiased under no carryover/time effects).

Estimator for the grand average τ: $\hat{\tau} = \bar{d} = \frac{1}{N}\sum_{i=1}^{N} d_i$.

5 Block Design

Units are partitioned into B blocks based on a pre-measured covariate. Within each block k, a completely randomized experiment is run independently: $N_t(k)$ of the $N(k)$ units in block k are assigned to treatment, the rest to control. Fisher's dictum: **block what you can, randomize what you cannot.**

Assignment Mechanism

$$|W^+| = \prod_{k=1}^{B} \binom{N(k)}{N_t(k)}, \qquad P(W) = \prod_{k=1}^{B} \binom{N(k)}{N_t(k)}^{-1}$$

The design factorizes as B independent CREs, one per block.

Fertilizer Example

Group plots into blocks by soil type (clay, loam, sandy). Within each block, randomly assign half the plots to the new fertilizer.

Key Insight

Blocking removes between-block variation from the error term. The variance of $\hat{\tau}$ depends only on within-block variation. The larger the between-block differences, the greater the precision gain over a CRE.

Three Levels of Causal Estimands

- Unit-level within block k: $\tau_i(k) = Y_i(1) - Y_i(0)$ for $i \in$ block k
- Block-average effect: $\tau(k) = \frac{1}{N(k)}\sum_{i\in k}\tau_i(k)$
- Population-weighted overall ATE: $\tau = \sum_{k=1}^{B}\frac{N(k)}{N}\tau(k)$

The weights $N(k)/N$ are population-size weights, not equal weights. Averaging $\tau(k)$ with equal weight would answer "what is the average effect across blocks?" rather than "across people?"

Fisherian Inference for Block Design

Sharp null: $H_0 : Y_i(1) = Y_i(0)$ for all i. The natural test statistic is the block-size-weighted average of within-block differences:

$$T_\lambda = \sum_{k=1}^{B} \lambda_k \big(\bar{Y}_T^{\text{obs}}(k) - \bar{Y}_C^{\text{obs}}(k)\big), \qquad \lambda_k = \frac{N(k)}{N}$$

> **Note**
>
> Re-randomize **only within blocks**: a unit in block k can never be reassigned to block k'. The reference set has $\prod_k \binom{N(k)}{N_t(k)}$ elements, not $\binom{N}{N_T}$. Using the wrong assignment space is a common mistake.

Neymanian Inference for Block Design

Block-level estimator: $\hat{\tau}(k) = \bar{Y}_T^{\text{obs}}(k) - \bar{Y}_C^{\text{obs}}(k)$, unbiased for $\tau(k)$.

Overall ATE estimator:

$$\hat{\tau} = \sum_{k=1}^{B} \frac{N(k)}{N}\, \hat{\tau}(k) = \sum_{k=1}^{B} \frac{N(k)}{N} \big(\bar{Y}_T^{\text{obs}}(k) - \bar{Y}_C^{\text{obs}}(k)\big)$$

Exact variance:

$$\text{Var}(\hat{\tau}) = \sum_{k=1}^{B} \left(\frac{N(k)}{N}\right)^2 \left(\frac{S_T^2(k)}{N_t(k)} + \frac{S_C^2(k)}{N(k) - N_t(k)} - \frac{S_{TC}^2(k)}{N(k)}\right)$$

where $S_T^2(k)$, $S_C^2(k)$ are finite-population variances within block k, and $S_{TC}^2(k) = \frac{1}{N(k)-1} \sum_{i \in k} (\tau_i(k) - \tau(k))^2$ is unobservable.

Conservative estimated variance (set $S_{TC}^2(k) = 0$):

$$\widehat{\text{Var}}(\hat{\tau}) = \sum_{k=1}^{B} \left(\frac{N(k)}{N}\right)^2 \left(\frac{s_T^2(k)}{N_t(k)} + \frac{s_C^2(k)}{N(k) - N_t(k)}\right)$$

Effect Modification

If there are two blocks $k = f$ (female) and $k = m$ (male), the treatment effect may differ across blocks. The estimand $\tau(m) - \tau(f)$ captures this heterogeneity. The unbiased estimator is $\hat{\tau}(m) - \hat{\tau}(f)$, with conservative estimated variance:

$$\widehat{\text{Var}}\big(\hat{\tau}(m) - \hat{\tau}(f)\big) = \frac{s_T^2(f)}{N_t(f)} + \frac{s_C^2(f)}{N(f) - N_t(f)} + \frac{s_T^2(m)}{N_t(m)} + \frac{s_C^2(m)}{N(m) - N_t(m)}$$

6 Factorial Design (2^K)

Study K binary factors simultaneously, each at two levels $(-1, +1)$. All 2^K treatment combinations are assigned to units.

Constructing the Lexicographic Order

The 2^K treatment combinations are listed in **lexicographic order**: the **first factor changes slowest** and the **last factor changes fastest**. Formally, factor k alternates in blocks of 2^{K-k} rows.

Note: Step-by-step construction

Step 1. Create a table with 2^K rows and K columns (one column per factor).

Step 2. Fill the **last column** (factor K): alternate $-1, +1, -1, +1, \ldots$ every single row.

Step 3. Fill column $K-1$: alternate in blocks of 2: $-1, -1, +1, +1, -1, -1, +1, +1, \ldots$

Step 4. In general, fill column k: repeat 2^{K-k} copies of -1 then 2^{K-k} copies of $+1$, cycling until all rows are filled.

Step 5. Fill the **first column** (factor 1): the first 2^{K-1} rows get -1, the last 2^{K-1} rows get $+1$.

Example ($K = 3$, factors A, B, C):

Run	A (block 4)	B (block 2)	C (block 1)
1	-1	-1	-1
2	-1	-1	$+1$
3	-1	$+1$	-1
4	-1	$+1$	$+1$
5	$+1$	-1	-1
6	$+1$	-1	$+1$
7	$+1$	$+1$	-1
8	$+1$	$+1$	$+1$

A flips once (after run 4). B flips every 2 runs. C flips every run.

Note: Checking your table

A correct lexicographic table satisfies:

- Each column is balanced: exactly 2^{K-1} entries of -1 and 2^{K-1} entries of $+1$.
- All 2^K rows are distinct (no duplicates).
- Any interaction column equals the elementwise product of its factor columns: e.g. column AB = column $A \times$ column B entry-by-entry. Use this to verify your work.

$K = 2$ Fertilizer Example

Factor 1: fertilizer dose (low $= -1$, high $= +1$). **Factor 2:** irrigation level (minimal $= -1$, intensive $= +1$). This gives $2^2 = 4$ treatment combinations applied to different plots.

Main Effects, Interactions, and Contrast Vectors

The **main effect** of factor k is the average outcome difference when factor k moves from -1 to $+1$, averaging over all combinations of other factors.

An **interaction** between j and k exists when the effect of j depends on the level of k. The 2^K design estimates all K main effects and $\binom{K}{2}$ two-way interactions jointly.

Each effect corresponds to a **contrast vector** $c \in \{-1, +1\}^{2^K}$:

- Main effect of factor k: c_j = level of factor k in combination j
- Interaction of j and k: $c = c_j \odot c_k$ (elementwise product)

Estimated effect: $\hat{\tau}_c = \frac{1}{2^{K-1}} \sum_{j=1}^{2^K} c_j \bar{Y}_j$, where $\bar{Y}_j$ is the mean outcome under combination j.

Fisherian Inference for Factorial Design

Note

To test whether effect c is zero under H_0:

$$T = \hat{\tau}_c = \frac{1}{2^{K-1}} \sum_{j=1}^{2^K} c_j \bar{Y}_j$$

Compute the p-value by re-permuting the assignment vector over all $|W^+|$ factorial assignments, recomputing T each time.

7 Design Summary

Design	$\lvert W^+ \rvert$	**Key feature**
CRE	$\binom{N}{N_T}$	Unbiased; possible covariate imbalance
Matched pair	2^P	Double index (i, p); CI exact via s_d^2/P
Multi-treatment	$N!/\prod N_j!$	J arms; ANOVA F-test; $\binom{J}{2}$ pairwise ATEs
Crossover	2^N	Self-control; d_i estimable; no carry-over required
Block	$\prod_k \binom{N(k)}{N_t(k)}$	Within-block randomization; size-weighted $\hat{\tau}$
Factorial (2^K)	2^K	Main effects + interactions via contrast vectors
Fractional (2^{K-p})	2^{K-p}	Fewer runs; aliasing defined by generators

Design determines: assignment space $|W^+|$, which covariates are balanced, the variance of $\hat{\tau}$, and the correct re-randomization space for Fisherian inference.

Fisherian: exact p-value; re-randomize within design constraints; no effect size.

Neymanian: $\hat{\tau} \pm$ CI; conservative because $S^2_{jj'}$ (or $S^2_{TC}(k)$, $S^2_{ct}(p)$) is unobservable; tightens under constant treatment effects or good pairing/blocking.